

# Case Study: Automated Multi-Industry B2B Lead Generation Engine

**Client Industry:** Digital Marketing & Sales Operations

**Target Market:** Small-to-Medium Businesses (SMBs) in the United States

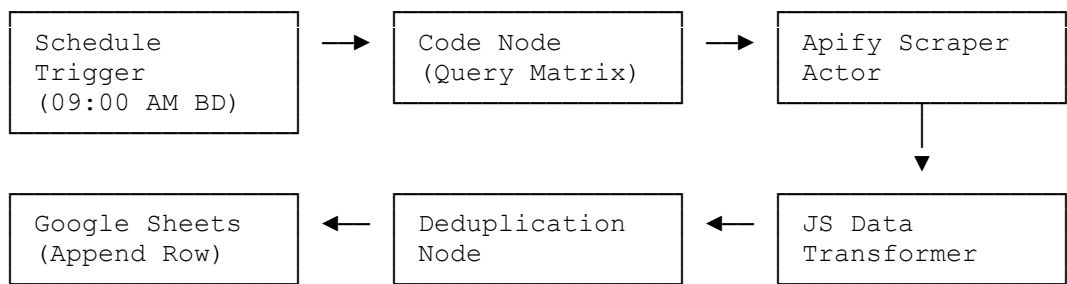
**Tech Stack:** n8n (Self-Hosted Architecture), Apify (Web Scraping Actors), Google Sheets API, JavaScript (ES6+), Docker, ngrok

## Executive Summary

A US-based digital marketing agency required a continuous pipeline of high-intent B2B leads across targeted local industries (Dental, Legal, Real Estate, Construction, and Accounting) with employee counts between 10 and 100. Manual research was inefficient, costly, and difficult to scale.

To solve this, an enterprise-grade automated lead generation pipeline was engineered using n8n and Apify. Operating on a scheduled cron trigger, the pipeline queries Google Maps and business directories, normalizes dirty data, strips duplicate entities across historical runs, and populates a client-facing spreadsheet. The system automatically extracts 100–150 qualified SMB leads daily without manual oversight.

## Technical Architecture & Pipeline Design



### Key Components & Data Flow

- Scheduled Triggering:** Configured for daily execution at 9:00 AM (Asia/Dhaka timezone) to match client reporting cycles.
- Dynamic Search Execution:** A custom JavaScript payload constructs targeted queries combining high-value industries with US geographical coordinates.
- Data Extraction via Apify:** Apify Actors scrape web listings, business registries, and Google Maps to capture raw operational data.

4. **Data Normalization & Cleaning:** Raw JSON payloads are sanitized via JavaScript to extract missing emails, parse direct phone numbers, and standardise operational fields.
5. **Cross-Execution Deduplication:** An n8n Deduplication node evaluates incoming domain properties (`Website`) against all historical workflow executions to prevent double-prospecting.
6. **Automated Storage:** Sanitized, unique lead records are appended directly into the client's production Google Sheet.

## Data Schema Specification

To maintain clean CRM integration, all extracted items are standardized into the following TSV/JSON schema before appending:

| Field Name     | Type   | Description                                      |
|----------------|--------|--------------------------------------------------|
| Business Name  | String | Legal trade name of the prospect business        |
| Website        | URL    | Primary canonical web address                    |
| Industry       | String | Categorized vertical (e.g., Dentist, Law Firm)   |
| Location       | String | Full physical address (Street, City, State, ZIP) |
| Phone          | String | Sanitized contact number                         |
| Email          | String | Public contact or support email address          |
| Contact Person | String | Identified executive or business owner           |

| Field Name   | Type   | Description                                   |
|--------------|--------|-----------------------------------------------|
| Job Title    | String | Title of the target contact                   |
| LinkedIn URL | URL    | Company or decision-maker social profile link |
| Company Size | String | Estimated employee tier (10–100 range)        |
| Source URL   | URL    | Original web source or Google Maps entity URL |

## Key Challenges & Solutions

### Challenge 1: Redundant Prospect Contacting Across Campaigns

- **Issue:** Running multi-keyword searches every morning generated recurring overlap for established local businesses.
- **Solution:** Integrated n8n's persistent "**Remove items processed in previous executions**" logic keyed on the unique `website` field. This guaranteed that previously contacted companies are filtered out automatically before reaching the CRM layer.

### Challenge 2: Local Self-Hosting & Runtime Reliability

- **Issue:** Self-hosting n8n locally behind an ngrok tunnel exposed the automated schedule to potential host downtime or tunnel disconnects.
- **Solution:** Configured free static ngrok domains (`ngrok http --domain=...`), aligned host sleep settings, and outlined dockerized deployment configurations on a VPS for true 24/7 cloud execution.

## Results & Impact

- **100% Workflow Automation:** Completely eliminated manual lead searching and data entry for sales staff.
- **Zero Duplication Rate:** 100% filter rate on recurring entities across historical runs.

- **Scalable Throughput:** Consistently delivers **100–150 qualified leads per day** direct to the client's Google Sheet ready for cold outreach.